

Applications of Bacterial Genomics: AMR surveillance

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What comes to your mind when you hear bacterial genomics?

- <https://www.menti.com/alokdehh1occ>



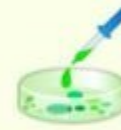
What comes to mind when you hear **Bacterial Genomics**?



DNA / Genes



Mutations & Evolution



Antimicrobial Resistance



Surveillance & Public Health



Phylogenetic Trees



Outbreak Investigation

Key terminologies

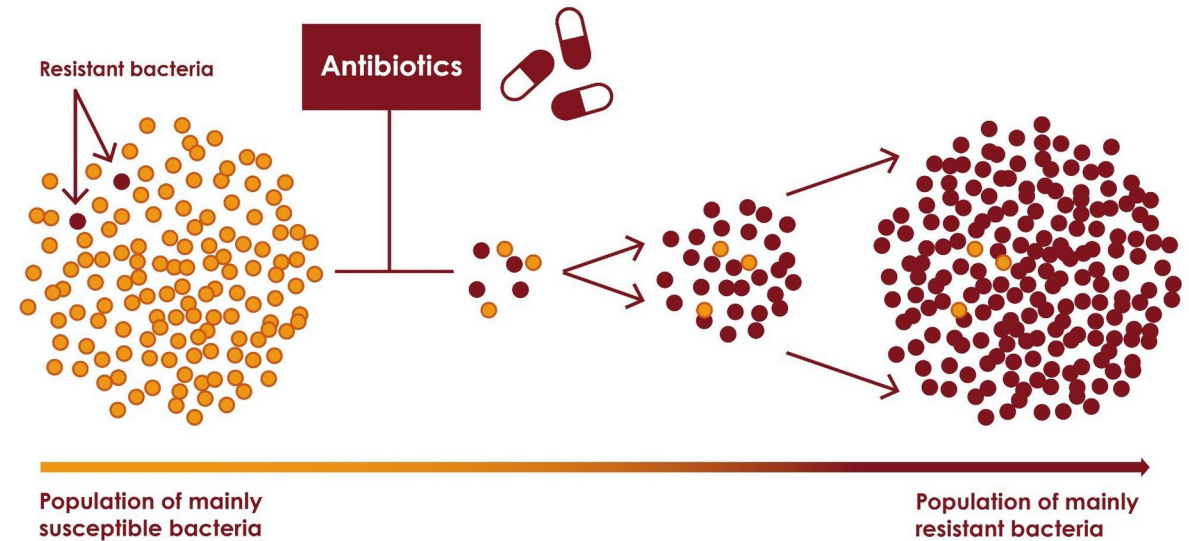
1. AMR (Antimicrobial Resistance)

- The ability of bacteria to survive drugs designed to kill them.
- Example: *E. coli* resistant to ciprofloxacin.

2. Mutations

- Changes in the DNA sequence of an organism.
- Some mutations cause resistance to antibiotics.

Natural selection of resistant bacteria



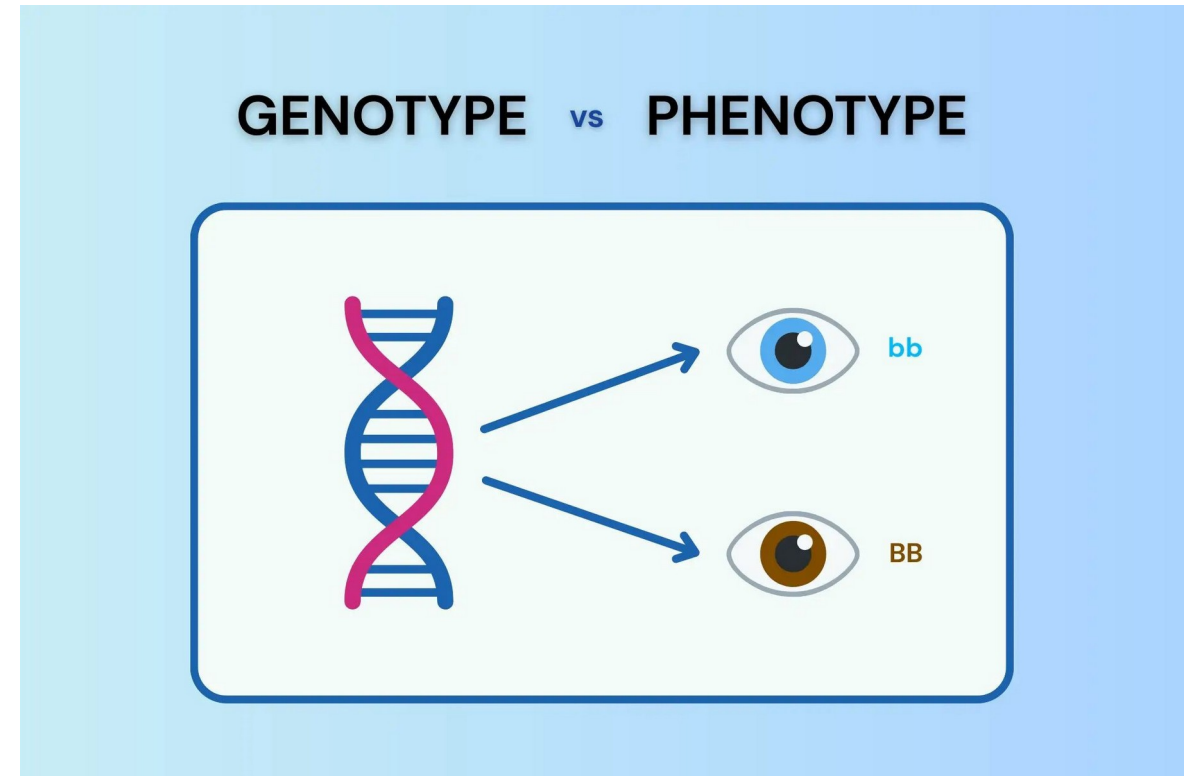
Key terminologies

3. Genotype

- The genetic makeup of an organism (the information in its DNA).
- Example: Presence of a resistance gene.

4. Phenotype

- The observable traits of an organism, influenced by its genotype and environment.
- Example: Bacterium growing in presence of antibiotics (resistant phenotype).



Key terminologies

5. Public Health

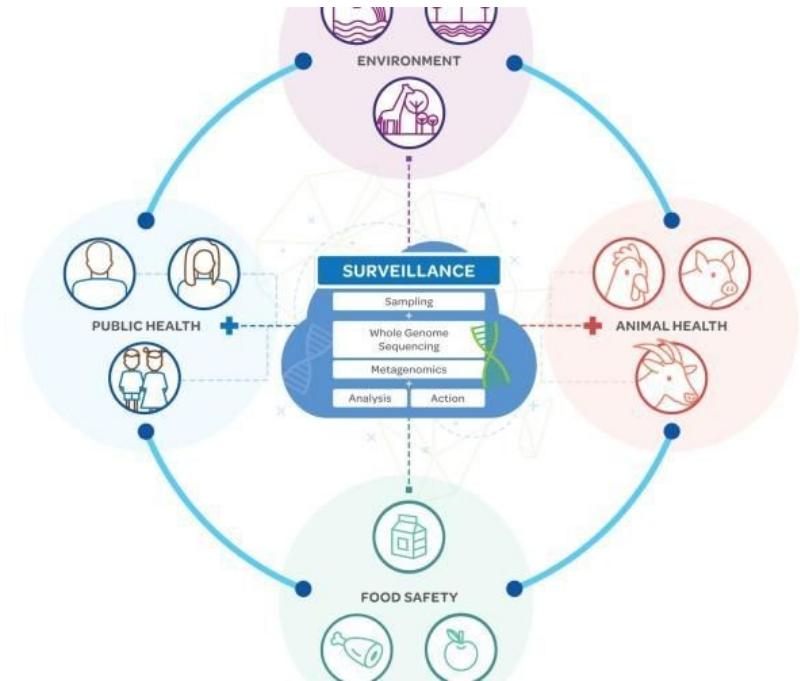
- Protecting and improving health of populations through prevention, surveillance, and interventions.

6. Surveillance

- Systematic monitoring of pathogens to track AMR, spread, and outbreaks.

7. Sequencing

- Determining the order of DNA bases (A, T, C, G) in a genome.
- Foundation of bacterial genomics.



Key terminologies

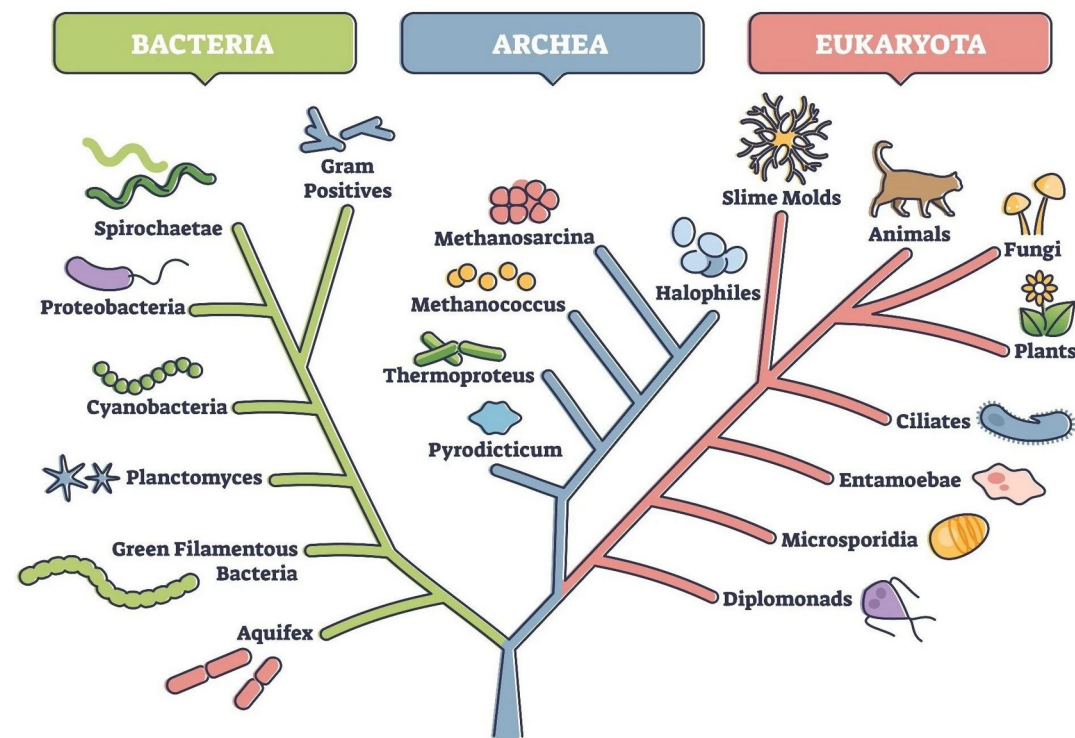
9. Database

- Organized collection of data that stores genomic, AMR, or epidemiological information.
- Examples: NCBI, CARD, ResFinder.

10. Phylogenetic Tree

- A branching diagram showing evolutionary relationships between organisms based on DNA sequences.
- Helps track bacterial spread and outbreaks.

PHYLOGENETIC TREE



Tracking Antimicrobial Resistance with Genomics



Tracking Antimicrobial Resistance

Detects AMR genes, mutations, and plasmid-mediated spread, guiding treatment decisions and stewardship.

Tracking Antimicrobial Resistance with Genomics

Gather

- Sequence isolate
- Assemble genome

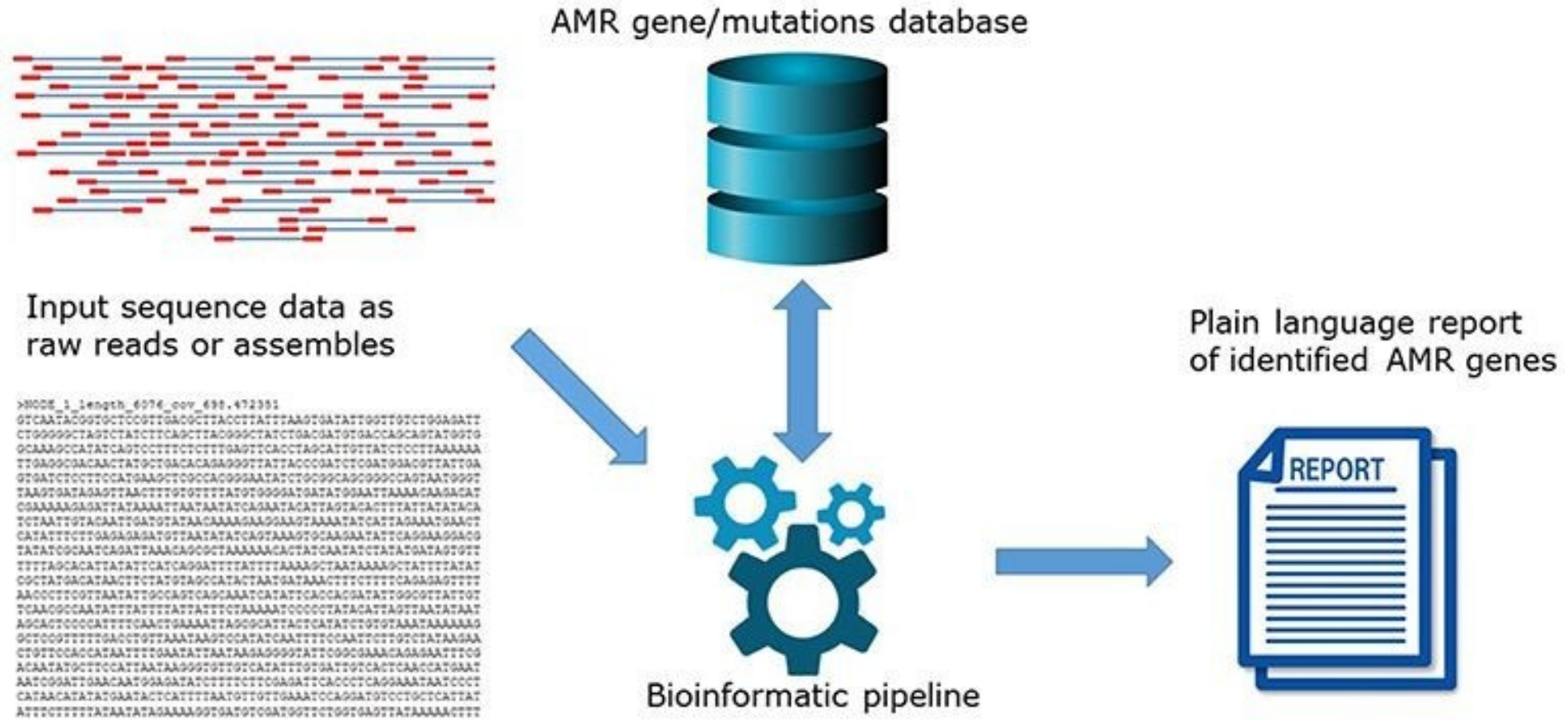
Evaluate

- Identify resistance genotype

Assess

- Compare to phenotype
- Look for new mechanisms

Tracking Antimicrobial Resistance with Genomics



Public Health Surveillance and Bacterial Genomics



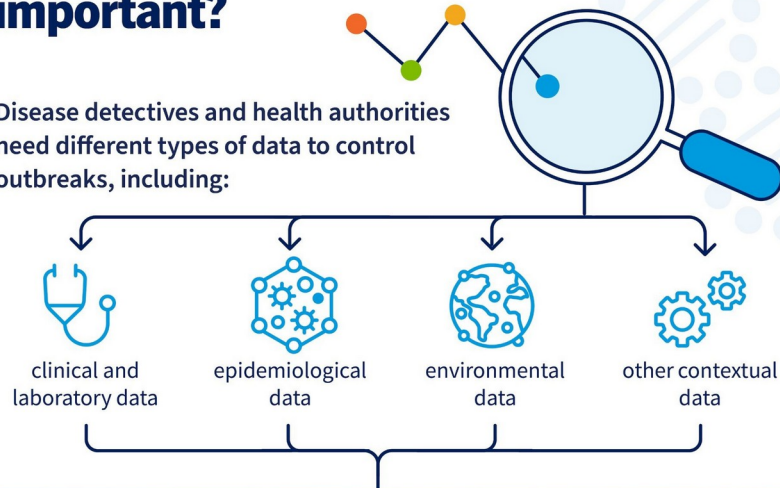
Public Health Surveillance

Monitors bacterial transmission in hospitals, communities, and the environment to inform outbreak response.

Public Health Surveillance and Bacterial Genomics

Why is GENOMIC SURVEILLANCE important?

Disease detectives and health authorities need different types of data to control outbreaks, including:

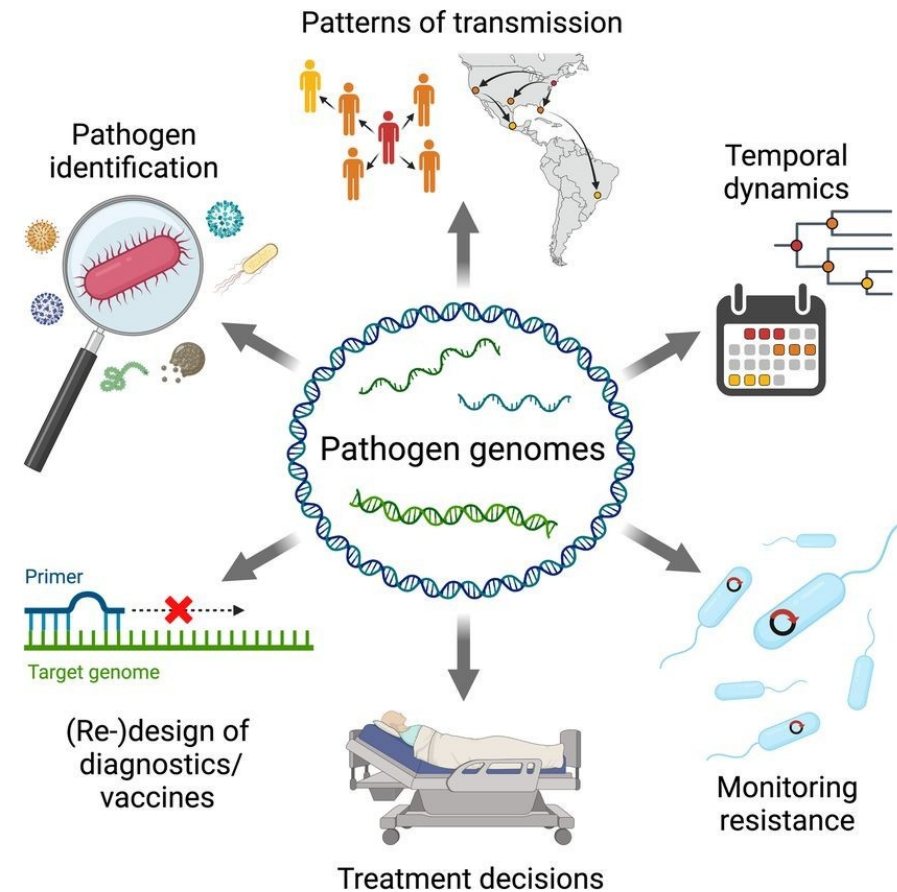


By adding genomic data, they can more quickly understand how a pathogen behaves and how to control it.



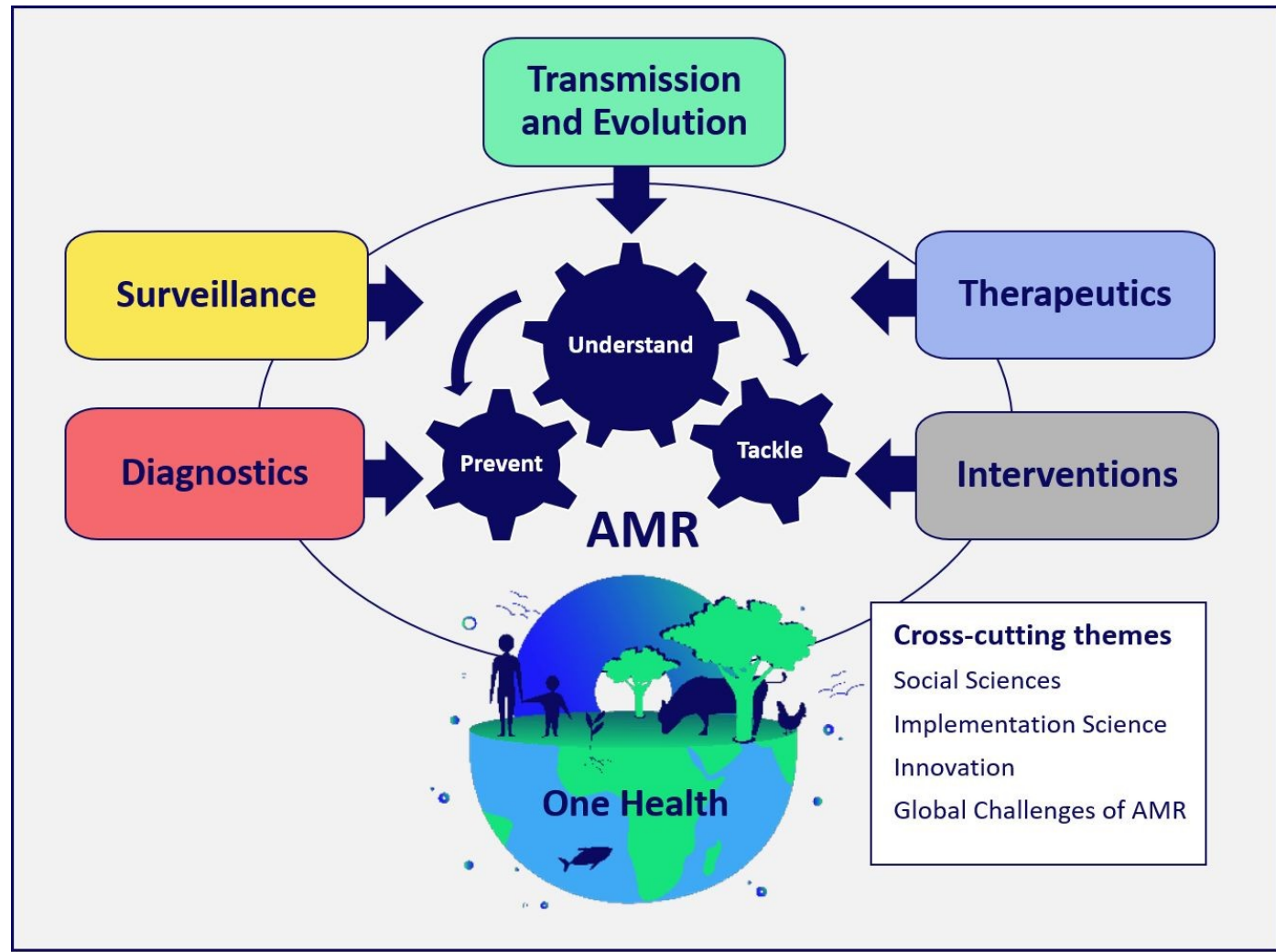
This is a powerful tool in public health surveillance.

The Global Genomic Surveillance Strategy
for Pathogens with Pandemic and Epidemic Potential



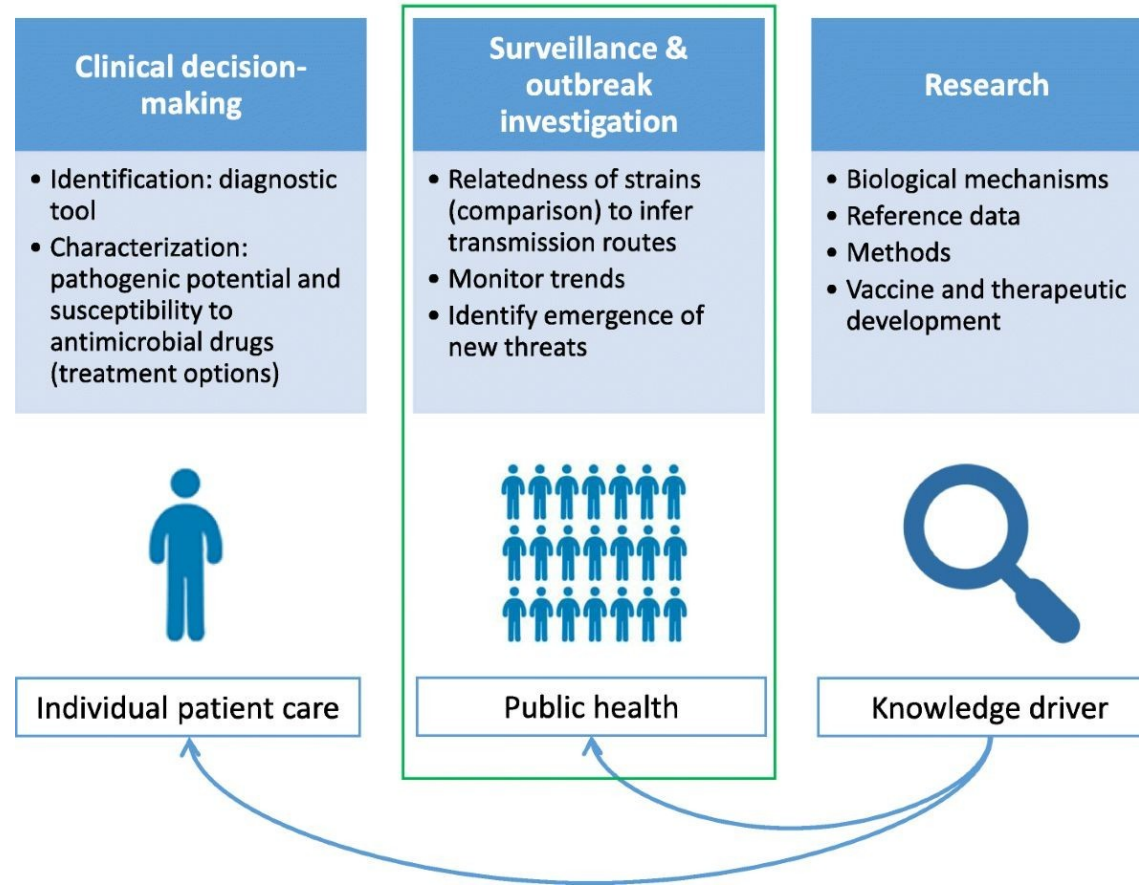
Why Surveillance Matters

- AMR is a One Health issue (humans, animals, environment)
- Requires monitoring across hospital, community, and environment
- Genomics strengthens surveillance capacity



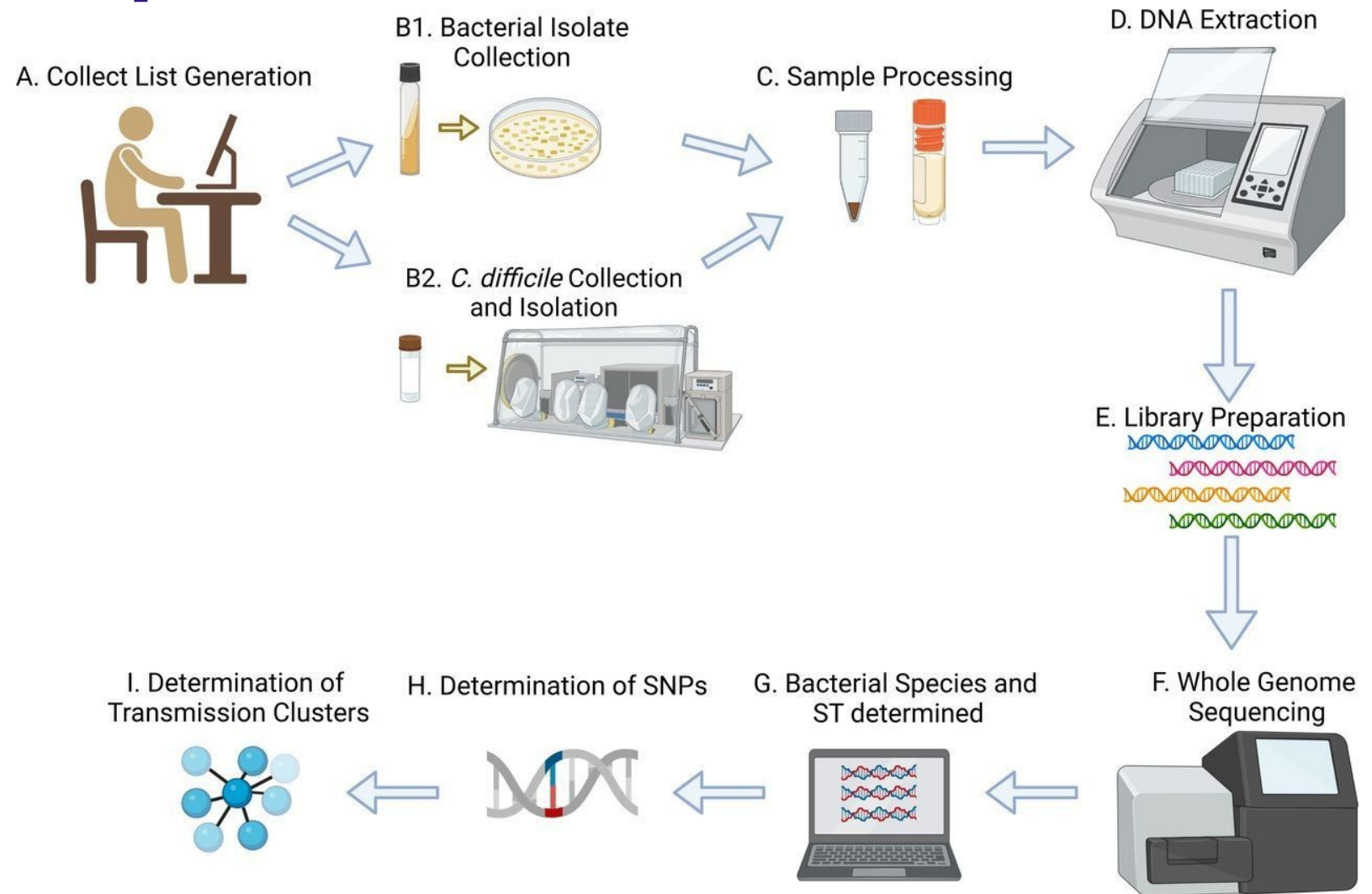
Public Health Surveillance and Bacterial Genomics

FOCUS

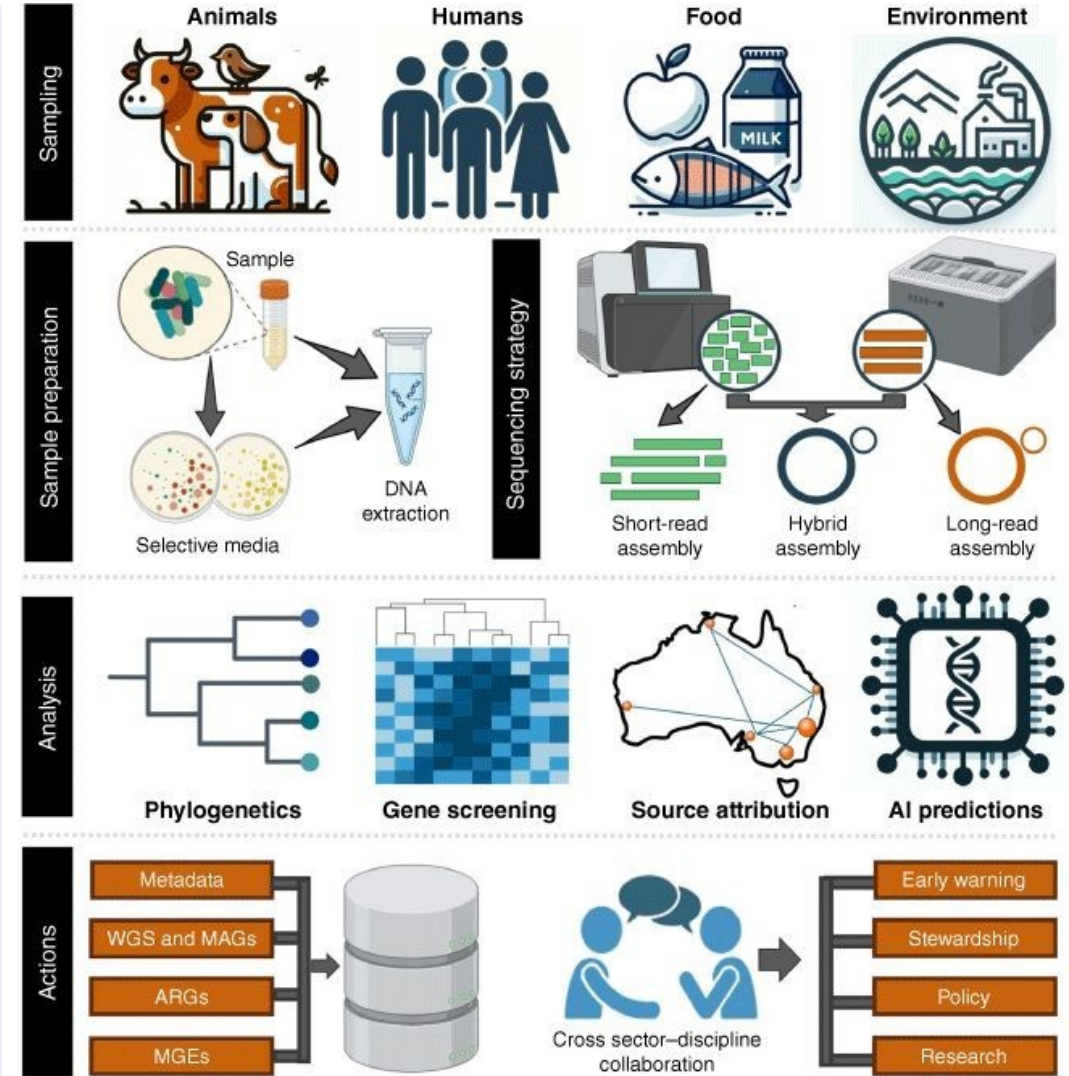
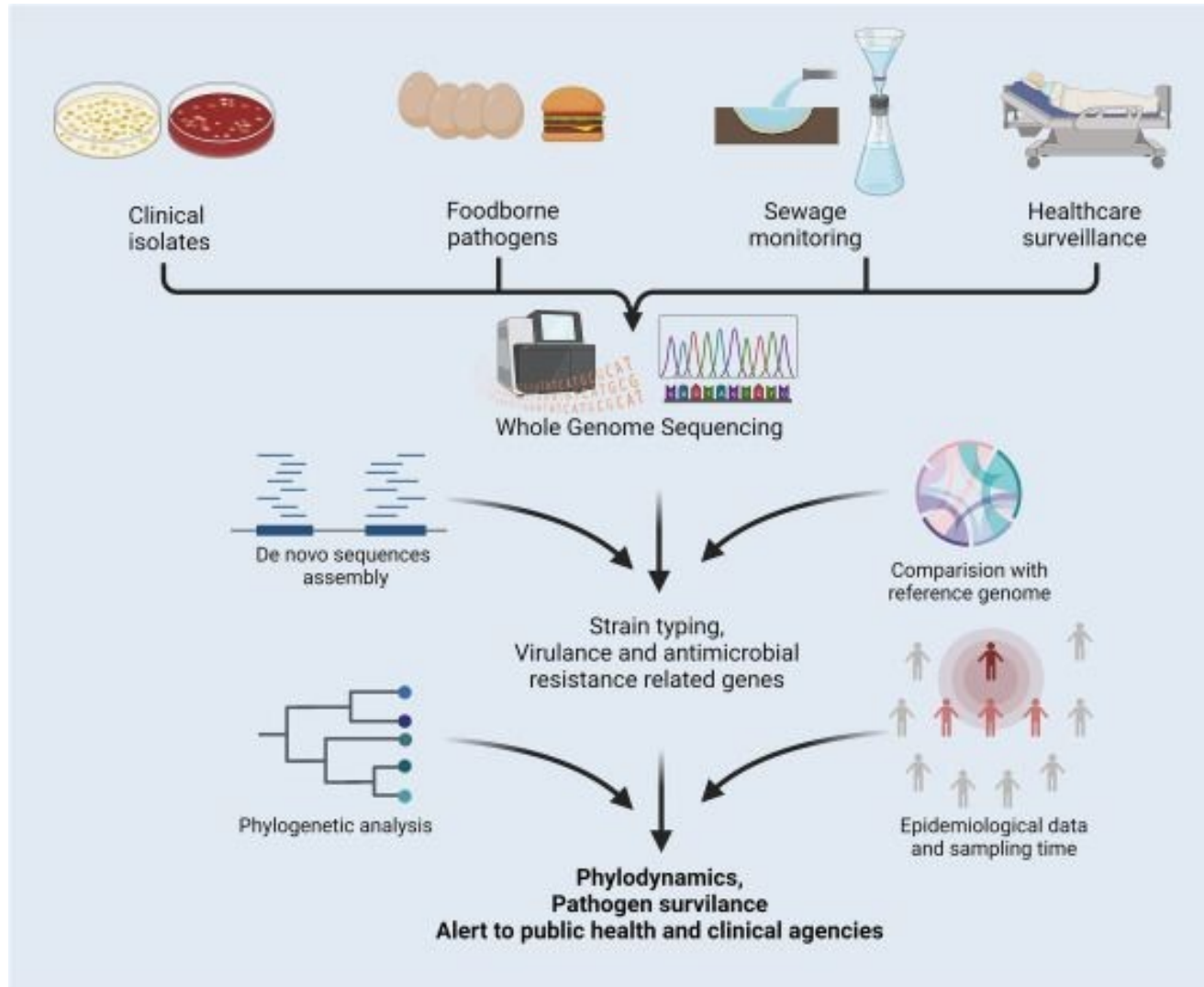


Hospital Surveillance

- Example: *Klebsiella pneumoniae* outbreaks in ICUs
- Genomics reveals patient-to-patient transmission
- Phylogenetic trees identify outbreak clusters



Community Surveillance



Interactive Discussion

Question:

- If your country wanted to strengthen AMR surveillance, which setting would you prioritize and why?

